Artificial Intelligence and Machine Learning - Answer Sheet - Student 04

Question 1: Explain the differences between supervised, unsupervised, and reinforcement learning in machine learning. Provide examples of applications for each approach.

Answer: Supervised learning is when the computer learns from examples with labels. Like if you show it pictures of cats and dogs and tell it which is which, then it can recognize new pictures. It's used for things like spam filters and predicting weather. Unsupervised learning doesn't have labels. The computer tries to find patterns on its own. It's like grouping similar things together without being told what the groups should be. This is used for customer segmentation and finding anomalies. Reinforcement learning is like training a dog with treats. The computer gets rewards when it does something right and penalties when it does something wrong. This is used for games and robots. The main difference is supervised has labels, unsupervised doesn't, and reinforcement uses rewards and punishments.

Question 2: Describe the architecture and functioning of Convolutional Neural Networks (CNNs) and explain why they are particularly effective for image recognition tasks.

Answer: CNNs are neural networks used for image recognition. They have convolutional layers that apply filters to images to detect features like edges and shapes. Then there are pooling layers that reduce the size of the data. Finally, there are fully connected layers that make the classification. They work well for images because they can detect patterns regardless of where they are in the image. They also use fewer parameters than regular neural networks, which makes them faster and less likely to overfit. CNNs are used for face recognition, object detection, and medical imaging. They're better than older methods because they automatically learn important features from the data instead of requiring humans to specify what to look for.

Question 3: Discuss the ethical considerations and potential societal impactsof implementing artificial intelligence systems in critical decision-making processes.

Answer: AI in decision-making has ethical problems. AI can be biased if the training data is biased. For example, if an AI for loan approval is trained on data where certain groups were unfairly denied loans, it will continue this discrimination. AI systems are often black boxes, so we don't know how they make decisions. This is a problem for important decisions like medical diagnoses or criminal sentencing. Privacy is another issue because AI needs lots of data, which might include personal information. There's a risk of surveillance and data breaches. AI might also replace jobs, causing unemployment. While some new jobs might be created, many people could be left without work. It's also unclear who is responsible when AI makes mistakes. Is it the developer, the user, or someone else? These problems need to be addressed with regulations and ethical guidelines.

Question 4: Explain the concept of transfer learning in deep neural networks and discuss its advantages and limitations.

Answer: Transfer learning is using a pre-trained model for a new task instead of starting from scratch. For example, you can take a model trained to recognize objects and adapt it to recognize medical conditions in X-rays. The advantages are: You need less training dat Training is faster Performance can be better The limitations are: It only works if the tasks are similar The original model might have biases Sometimes you need to retrain the whole model anywayTransfer learning is popular because not everyone has the resources to train models from scratch.

Question 5: Describe the principles of natural language processing (NLP) and how transformer-based models like BERT have revolutionized language understanding tasks.

Answer: NLP is about making computers understand human language. Traditional NLP used rules and statistics to process text, but had limitations in understanding context and meaning. Transformer models like BERT changed NLP by using attention mechanisms that look at all words in a sentence at once, instead of processing one word at a time. This helps them understand context better. BERT is trained on huge amounts of text data to predict missing words and

understand relationships between sentences. After this pre-training, it can be fine tuned for specific tasks like question answering or sentiment analysis. These models are better because they understand context and can handle different meanings of the same word. They've improved performance on many language tasks, but they still have problems with bias and require a lot of computing power.